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T11 – 05

Assignment No. 10

Aim: Analyse the PWA using the Google Lighthouse PWA Analysis Tool.

Theory:

What is Google Lighthouse?

Google Lighthouse is an open-source, automated tool provided by Google for improving the quality of web pages. It can audit performance, accessibility, SEO, best practices, and most importantly—Progressive Web App (PWA) capabilities.

Lighthouse runs directly in the Google Chrome DevTools and generates a detailed report with scores and improvement suggestions.

Purpose of Using Lighthouse for PWA

Lighthouse helps developers:

- Verify if their web app qualifies as a PWA
- Test installability, offline support, and responsiveness
- Ensure the app meets PWA best practices

How to Use Lighthouse to Analyze a PWA

1. Run Your Web App

Start your application on a local server (e.g., <http://localhost:3000>) or use a live deployed version.

2. Open Chrome DevTools

- Right-click → Inspect, or press Ctrl+Shift+I

- Go to the “Lighthouse” tab (use the » icon if it's hidden)

3. Choose Audit Options

- Select “Progressive Web App” under categories
- Choose device: Mobile or Desktop
- Click on “Analyze page load” or “Generate report”

4. Review the Report

Lighthouse will generate a detailed PWA report showing:

- Installability check (manifest + icons)
- Service worker and offline capabilities
- Performance and load speed
- Recommendations for improvements



Key Metrics Checked in a PWA Audit

Metric	What it Tests
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Manifest Validity	Proper manifest.json file, icons, app name, etc.
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Service Worker	Registered and caching files for offline use
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HTTPS	Secure connection required for PWA
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Offline Support	App should work without internet
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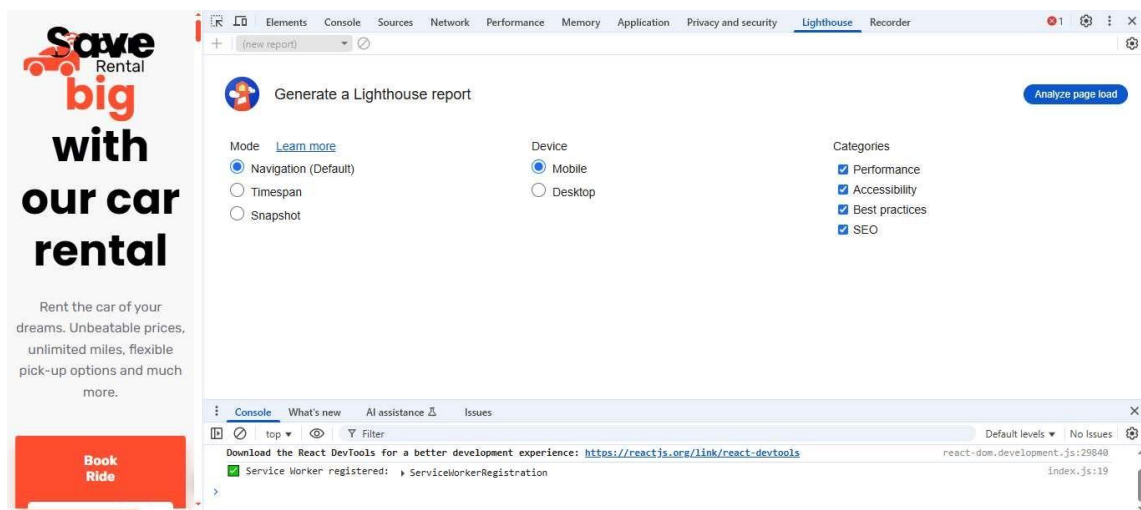
Responsive Design	Mobile-first and adaptable to different screen sizes
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Conclusion

Google Lighthouse is an essential tool for evaluating and improving PWA readiness. By analyzing your web app with Lighthouse, you ensure it meets the technical standards of a Progressive Web App, providing a reliable, fast, and applike experience to users.

Output:





There were issues affecting this run of Lighthouse:

- Chrome extensions negatively affected this page's load performance. Try auditing the page in incognito mode or from a Chrome profile without extensions.



Performance

